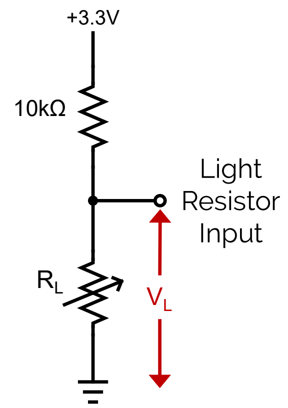
Recommended Assessment

Light Resistors

Questions

1. What happens when you move a dark object closer to or farther from the light resistor? What happens when you move a sheet of white paper closer to or farther from the light resistor?

2. Using the following circuit diagram that shows the internal circuit for the light resistor input, explain how the voltage across the light resistor changes with the amount of light detected. Does the resistance increase or decrease when more light is detected?

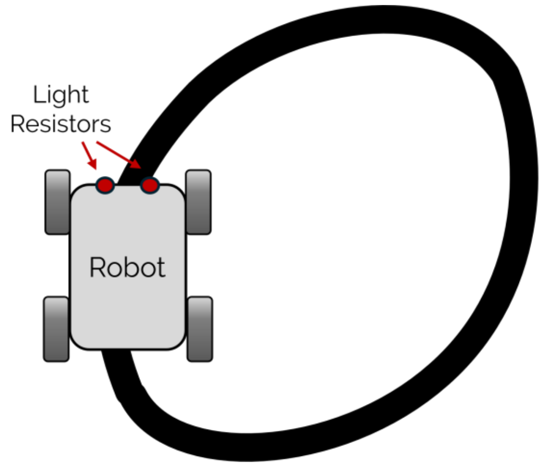


3. Referring to the datasheet, verify if the resistance increases of decrease when more light is detected. Report the value of the dark resistance of the light resistor. Use your equation from Question 2 to find the expected voltage at the dark resistance. How does this voltage compare with the measured voltage when you completely cover the light resistor?

4. How does the lighting of the environment affect the reading of the sensor? How does this affect the use of light resistors for detecting changes in light?

5. How does the orientation of the Mechatronic Sensors Trainer affect the amount of light detected by the sensor?

6. One application of light resistors is in simple line following robots. These robots use at least 2 light resistors, positioned facing down, to keep the robot centered as it drives along a black line on a white surface. The light resistors are usually placed on either side of the line, as shown in the figure below:



    a. Considering what happened when a dark object or a white piece of paper was placed in front of the sensor, explain how you would use light resistors for steering control in a line following robot.

    b. Considering what happened when you tested the light resistor in different environments, what would be the benefit of adding LEDs that point down?

Light Resistor Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	